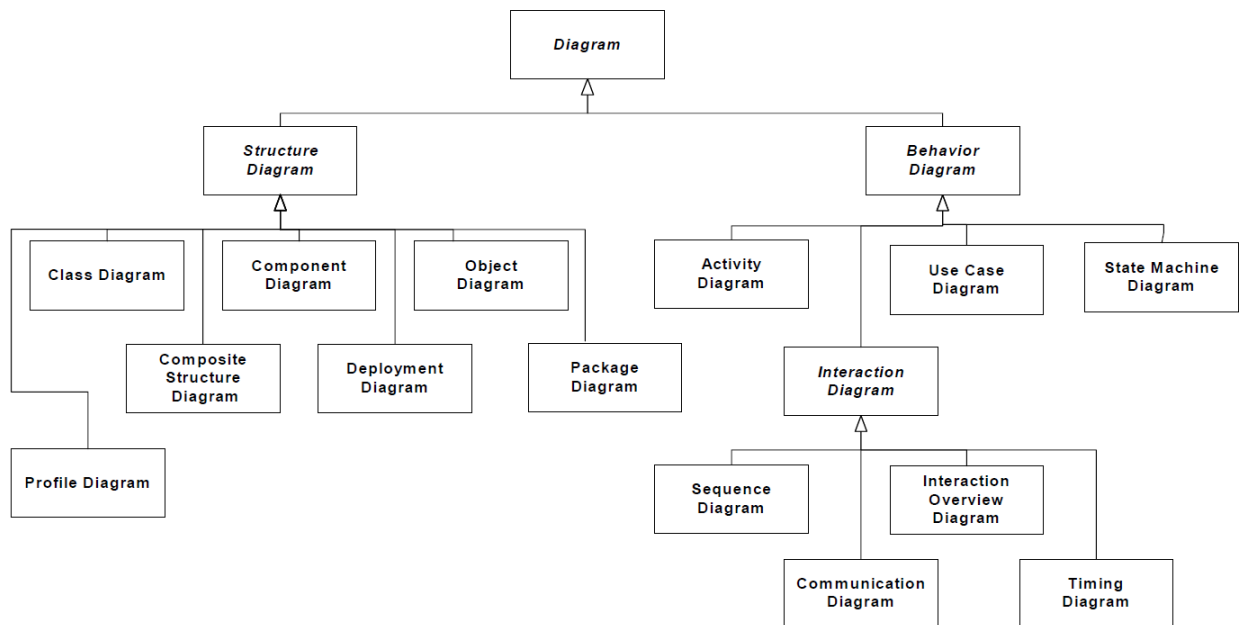


CSCI 578 – Software Architectures
Fall 2025
Individual Assignment #3
Due: Thursday, November 6, 2025

In the previous assignment, you developed a UML model(s) of your **PocketLLM Portal** app. However, that model was *independent* of any frontend framework. In this assignment, you will adapt your model to rely on a particular framework of your choice (e.g., React, Angular, Vue, Django, Rails, Spring, ASP.NET, etc.). You are still expected to only use UML. The UML specification can be found at: <https://www.omg.org/spec/UML/2.4.1/Superstructure/PDF>. It depicts the taxonomy of UML diagrams as follows (pages 693-694):



In updating your architecture to use the chosen framework, you should refine your model to include any components and any relevant attributes of all components that the framework requires. Put yourself in the shoes of a potential reader (e.g., a software developer) who will implement your PocketLLM Portal app according to your model(s). The more design decisions omitted from the model(s), the more assumptions the developer will have to make.

You should document any departures from your model(s) produced in Assignment #2 and the rationale behind each departure. You should also explain how and where your model had to change to comply with your chosen framework.

You will be evaluated on (1) your model's completeness,¹ (2) its compliance with the framework of choice, and (3) the (plain language) rationale behind all major architectural decisions you made. Like Assignment #2, you may use as many UML diagrams as needed to describe your

¹ Regarding "completeness," refer to the corresponding note from the Analysis of Software Architecture lecture.

architecture. As in earlier assignments, carefully review this document and ensure that every requirement has been fully addressed.

You must make your model **as if you will implement the PocketLLM Portal app according to the model(s)**. In other words, this is your opportunity to familiarize yourself with the framework of choice and to lay the foundation for implementing your PocketLLM Portal. Note that the team project will include implementation tasks based on your model(s) (details will be provided later).

You must also address any comments you received for Assignment #2. You should update your model accordingly if the comment was about the architecture. Any comments that have not been addressed will result in the exact deduction you received for Assignment #2.

Frameworks

Since the choice in the frontend frameworks for this assignment is open, make a reasonable assumption that the teaching staff will not be familiar with your chosen framework. Do your diligence and get familiar with the requirements and restrictions that the framework comes with.